



# Build a Security Monitoring System



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Alarms (1)

Q Search

Alarm state: Any

Alarm type: Any

Actions status: Any

< 1 >

Name

State

Last state update (Local)

Conditions

Actions

[Secret is accessed](#)

In alarm

2026-01-25 09:55:05

Secret is accessed >= 1 for 1 datapoints within 1 minute

Actions enabled



# Introducing Today's Project!

In this project, I will demonstrate how to build a security monitoring system in aws

## Tools and concepts

Services I learned were AWS Cloudtrail, AWS Secrets Manager, AWS Cloudwatch, AWS SNS.

## Project reflection

This project took me approximately 1 hour. The challenging part was figuring out why the SNS notification was not coming to my email.



# Create a Secret

Secrets Manager is a resource that helps us protect secrets, which are passwords, API keys, credentials and sensitive information. Instead of storing important credentials in the code or sharing them via email we can store them safely away in Secrets Manager.

To set up for my project, I created a secret called topsecretinfo which contains the confidential value.

## TopSecretInfo

### Secret details

Encryption key

aws/secretsmanager

Secret name

TopSecretInfo

Secret ARN

arn:aws:secretsmanager:ap-south-1:582577265973:secret:TopSecretInfo-OQzkRi

Secret description

Secret created for NextWork's project on Building a Monitoring System

Secret type

-



Actions ▾

Overview

Rotation

Versions

Replication

Tags



# Set Up CloudTrail

AWS CloudTrail is a monitoring service - think of it as an activity recorder throughout your AWS account. It documents every action taken, like who did what, when they did it, and where they did it from. A trail tells CloudTrail exactly what activity to record and where to save those recordings.

There are different types of CloudTrail events such as :- Management events: These show us admin actions that configure the AWS resources - creating an EC2 instance, updating a security group, or in our case, accessing a secret. Data events: These track high-volume actions that operate ON the resources rather than creating or configuring them - like uploading a file to S3 or running a Lambda function. Insights events: These detect unusual patterns in our management events, like someone suddenly creating 100x more IAM users than normal. Network activity events: These track network-related activities, like changes to your VPC configuration or traffic to a subnet.

## Read vs Write Activity

Read API activity happens when someone views but doesn't change anything. For example, listing S3 buckets, describing EC2 instances . Write API activity occurs when changes happen - creating, deleting, modifying resources, or even retrieving the value of a secret .



# Verifying CloudTrail

I retrieved the secret in two ways: First through the secrets manager console and second through the AWS Command-Line-Interface (CLI)

To analyze my CloudTrail events, I visited the event history tab in cloud trail and then with the help of lookup attributes i filtered the logs by the event source which points towards secretsmanager and i found a log with the eventname GetSecretValue which meant my secret value was used or retrieved

```
CloudShell
ap-south-1 +
~ $ aws secretsmanager get-secret-value --secret-id "TopSecretInfo" --region ap-south-1
{
  "ARN": "arn:aws:secretsmanager:ap-south-1:582577265973:secret:TopSecretInfo-QQzkRi",
  "Name": "TopSecretInfo",
  "VersionId": "89f19bd3-3928-4942-a306-2f075a9492c6",
  "SecretString": "{\"The Secret is\":\"rice is the best carb\"}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2026-01-25T05:19:20.919000+00:00"
}
~ $
```



# CloudWatch Metrics

Amazon CloudWatch Logs is a service that helps in bringing together logs from different AWS services, including CloudTrail, for visibility, troubleshooting, and analysis. It is important because once logs are in CloudWatch, we can create alerts based on specific patterns (such as someone accessing your secret), visualize trends, or trigger automated responses.

CloudTrail Event History, which is great for quick investigations on recent events. CloudWatch Logs is where we can set up alerts and automated responses when specific events happen. CloudTrail Event History only keeps events for 90 days, while CloudWatch Logs can store them for as long as you'd like. CloudWatch Logs has powerful filtering tools that let us focus on exactly the events we care about.

Metric value is what gets recorded when our filter spots a match in the logs. We're setting it to 1 so that each time someone accesses our secret, the counter increases by exactly one. Default value is what gets recorded when our filter doesn't find any matches during a given time period. We're setting it to 0 so that time periods with no secret access show up as zero on our charts, rather than not showing up at all. This gives us a complete picture - we can see both when access happened AND when it didn't.



### Metric details

#### Metric namespace

Namespaces let you group similar metrics. [Learn more](#) 

 Create new

Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.

#### Metric name

Metric name identifies this metric, and must be unique within the namespace. [Learn more](#) 

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(\*), dollar(\$), and space( ).

#### Metric value

Metric value is the value published to the metric name when a Filter Pattern match occurs.

Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. \$requestSize for delimited filter pattern or \$.status for JSON-based filter pattern - dollar (\$) or dollar dot (\$.) followed by alphanumeric and/or underscore (\_) characters).

#### Default value – *optional*

The default value is published to the metric when the pattern does not match. If you leave this blank, no value is published when there is no match. [Learn more](#) 

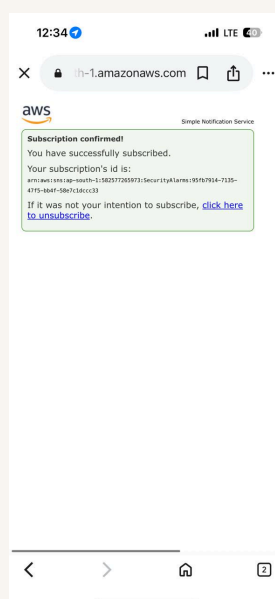


# CloudWatch Alarm

The alarm threshold is when the alarm should trigger. I set my CloudWatch alarm threshold to 1 so the alarm will trigger when the `SecretsAccessed` metric is greater than or equal to 1 in a 5-minute period.

I created an SNS topic along the way. An SNS (Simple Notification Service) topic is like a broadcast channel for your notifications. First, you create the channel (topic), then you invite subscribers (such as your email), and finally, you send messages to the topic. SNS automatically delivers that message to all subscribers.

AWS requires email confirmation because they want to make sure it's really me asking for these alerts.







# Troubleshooting Notification Errors

To test my monitoring system, I tried accessing secrets again but i did not get any mail from AWS.

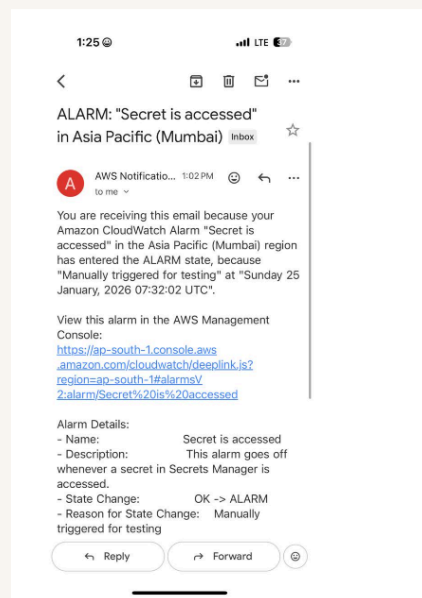
When troubleshooting the notification issues I first checked in cloutrail if my logs are coming and then i checked whether the cloudtrail logs are going into cloudwatch or not and then i checked the metrics in cloudwatch and i got the issue.

I went to the cloud watch's metric section and there i got to know that the alarm condition i had specified was wrong. I had earlier specified greater than but it was supposed to be greater than or equal to. Once i changed the option to greater than or equal to i got an email from AWS notifications.



## Success!

To validate i checked the cloudtrail logs, then the cloudwatch logs and then i verified everything by the mail i got from AWS when i accessed my secret





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